MIL-C-13602C
30 October 1967
SUPERSEDING
MIL-C-13602 B
7 June 1961

MILITARY SPECIFICATION

CASE, PANEL MARKER FOR AERIAL COMMUNICATION PANEL MARKER SETS AP-30-C and AP-30-D

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers one type and size of cotton duck case used to house the AP-30-C and the AP-30-D panel marker sets.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-285	- Thread, Polyester
CCC-C-419	- Cloth, Duck, Cotton, Unbleached, Plied Yarns (Army and Numbered)
CCC-D-950	- Dyeing and Aftertreating Processes for Cotton Cloths
DDD-L-20	- Label; For Clothing, Equipage, and Tentage (General Use)
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	- Box, Fiberboard

FSC 8345

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MILITARY

MIL-F-495	-	Finish, Chemical, Black, for Copper Alloys
MIL-F-496	-	Clips, End
MIL-W-530	-	Webbing and Tape, Textile, Cotton, General Purpose, Natural or in Color
MIL-B-543	-	Buckles, Tongueless and Web Strap
MIL-F-10884	-	Fasteners, Snap

STANDARDS

FEDERAL

Fed. Std. No. 751	-	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

DRAWINGS

THE INSTITUTE OF HERALDRY

5-1-334	-	Case, Panel Marker for Aerial Communication Panel Marker Sets AP-30-C and AP-30-D, Assembly and Illustration
5-1-335	-	Case, Panel Marker for Aerial Communication Panel Marker Sets AP-30-C and AP-30-D, Sections
5-1-331	-	Case, Panel Marker for Aerial Communication Panel Marker Sets AP-30-C and AP-30-D; Details

(Figures 1, 2 and 3 are miniature reproductions of the referenced drawings and are attached for information only.)

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

COLOR ASSOCIATION OF THE UNITED STATES, INC.

Standard Color Card of
OFFICIAL STANDARDIZED SHADES OF SEWING THREADS,
U.S. ARMY

(Application for copies of the above color card should be addressed to the Color Association of the United States, Incorporated, 200 Madison Avenue, New York, New York 10016.)

3. REQUIREMENTS

3.1 Standard samples.- Unless otherwise specified (see 6.2 and 6.3), a standard sample of the cotton duck, tape, webbing and thread shall be furnished for matching color, shade and when specified fastness requirements.

3.2 Preproduction sample.- Unless otherwise specified (see 6.2), before production is commenced a sample of the finished case shall be submitted or made available to the contracting officer or his authorized representative for inspection and tests as specified in 4.2. The approval of the preproduction sample authorizes the commencement of production but does not relieve the contractor of responsibility for compliance with the provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.3 Materials.- Materials shall conform to the referenced specifications and the requirements specified herein.

3.3.1 Cloth, cotton duck.- The cotton duck before finishing shall conform to type I, No. 6 or No. 10, hard texture of CCC-C-419.

3.3.1.1 Color and treatment.- The duck shall be dyed to match the standard shade of O.D. No. 7 (see 3.1) and treated in accordance with type I, class B of CCC-D-950.

3.3.2 Tape, textile cotton.- The tape for the binding shall be 1 inch wide, dyed to match O.D. No. 7 (see 3.1) and shall conform to type I, class 4 of MIL-W-530.

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3.3.3 Webbing.- The webbing for the straps shall be 1 inch wide, dyed to match O.D. No. 7 (see 3.1) and shall conform to type III, class 4 of MIL-W-530.

3.3.4 Thread, polyester.- The thread for all stitching shall conform to type I or II, class 1, subclass a, size FF of V-T-285. The thread shall be shade S-1 (Cable No. 66022) and shall show fastness to chlorine bleaching equal to or better than the standard sample (see 3.1). When no standard sample is available, the thread shall show good fastness to chlorine bleaching.

3.3.5 Buckles.- The buckles for the straps shall be 1 inch and shall conform to type I, class 1, construction A of MIL-B-543.

3.3.6 Clip, end.- The end clip shall be brass, 1 inch conforming to type I, class 1 of MIL-C-496. The end clips shall have a black chemical finish in accordance with MIL-F-495.

3.3.7 Snap fasteners.- The snap fasteners shall conform to style 2, finish 2 of MIL-F-10884.

3.4 Construction.- The cases shall be constructed in accordance with drawings 5-1-334, 5-1-335 and 5-1-331 and the requirements specified herein. As an alternate method of construction, reinforcements numbered 4 and 5 as shown in section A-A and H-H of drawing 5-1-335 may be made in two pieces provided the butted edges are caught in the center stitching.

3.4.1 Stitching.- Stitching shall be in accordance with Fed. Std. No. 751. All stitching shall be accomplished using stitch type 301 with 6 to 8 stitches to the inch. All thread breaks shall be back stitched not less than 1 inch at each break. Except where stitching ends are turned under in a hem or caught by other stitching, stitching ends shall be back stitched not less than 1 inch. All thread ends shall trimmed. Thread tension shall be maintained so that there will be no loose stitching and so that the lock will be embedded in the center of the materials sewed.

3.4.2 Setting of end clips.- The webbing shall be inserted the full depth of the clip and the clip securely attached without damaging or cutting the webbing. The webbing shall be flat at the point of entrance.

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3.4.3 Setting of snap fasteners.- A hole shall be punched through the material smaller than the outside diameter of the fastener tube. The tube shall then be forced through the hole and the fastener securely clinched without damaging or cutting the material.

3.5 Color matching.- Color shade shall be matched to the standard under natural (North Sky) daylight or artificial daylight having a color temperature of 7500 degrees Kelvin and shall be a good approximation to the standard under incandescent lamplight at 2800 degrees Kelvin.

3.6 Marking.- Marking shall conform to the requirements specified for type IV, class 5 labels of DDD-L-20. In addition, the letters "U S" and the nomenclature of the item shall be applied in the location and characters specified on the applicable drawing and shall conform to the requirements specified for type IV, class 9 labels of DDD-L-20. Fastness shall be as specified for class 5 labels.

3.7 Workmanship.- All cases shall be clean, well made and shall meet the quality of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 Preproduction sample inspection.- Inspection of the preproduction sample shall be made on a completely fabricated case for all provisions of this specification applicable to end product examination.

4.3 Sampling for inspection and acceptance.- Sampling for inspection and acceptance shall be made in accordance with the provisions set forth in MIL-STD-105 except where otherwise specified.

4.3.1 Component and material testing.- In addition to the quality assurance provisions of subsidiary specifications and drawings components and materials shall conform to the requirements specified herein. In the event of a conflict this specification shall govern.

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4.3.2 In-process inspection.- In-process inspection shall be made at any point or during any phase of the manufacturing process to determine whether construction details which can not be examined in the finished product are in accordance with requirements indicated in the specification and drawings. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated non-conformance.

4.3.3 Inspection of the end item.-

4.3.3.1 Visual examination.- Defects found during visual examination shall be classified in accordance with 4.3.3.1.1 and 4.3.3.1.2. The lots shall be selected in accordance with 4.3.3.1.3. The unit of product for these examinations shall be one finished case.

4.3.3.1.1 Examination of cases for defects in material, design, construction and workmanship.- Defects designated by an (*) asterisk shall be considered major when seriously affecting appearance or serviceability and minor when affecting appearance or serviceability but not seriously.

Examine	Defects	Classification		
		Major	(*)	Minor
Fabric	Holes, cuts or tears.....	x	-	-
	Other fabric defects affecting serviceability or appearance...	-	*	-
Webbing	Not firmly or tightly woven; edges frayed or scalloped.....	x	-	-
Hardware	Broken or malformed; failing to serve intended purpose; any corroded areas; burrs or sharp edges which may cause injury in handling.....	x	-	-
	Finish totally or partially omitted.....	-	*	-
Seams and stitching	More than 1/4 inch, but not more than 1 inch.....	-	-	X
Open seams	More than 1 inch..... NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more continuous skipped stitches or run-offs occur. On double stitched seams, a seam shall be considered open when either one or both sides of the seam are open.	x	-	-

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Examine	Defects	Classification		
		Major	(*)	Minor
Raw edges (when securely caught in other seams or stitching)	More than 1/2 inch, but not more than 1 inch.....	-	-	x
	More than 1 inch.....	x	-	-
	NOTE: Raw edges not securely caught in stitching shall be classified as open seams.			
Run-offs (see open seams)				
Seam and stitch type	Wrong seam or stitch type.....	x	-	-
Stitch tension	Loose, resulting in a loosely exposed top or bobbin thread; tight, resulting in excessive tightness of fabric on seams...	-	*	-
Stitches per inch	One stitch less than minimum specified.....	-	-	x
	Two or more stitches less than minimum specified.....	-	*	-
	One or more stitches in excess of maximum specified.....	-	-	x
Stitching margins: Gage on double stitching	Not as specified, affecting serviceability.....	x	-	-
Thread breaks	Backstitched less than one inch.	-	-	x
	NOTE: Thread breaks not backstitched shall be classified as open seams.			
Stitching ends	Ends of stitching backstitched less than one inch (except where ends are turned under in a hem or held down by other stitching).....	-	-	x
Cleanness	Grease or oil stains clearly noticeable; thread ends not trimmed throughout.....	-	*	-
Marking	Omitted, misplaced, illegible, incorrect or size of characters not as specified.....	x	-	-

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Examine	Defects	Classification		
		Major	(*)	Minor
Components and assembly	Any required component or operation omitted (unless otherwise classified herein)..	x	-	-
	Mends, darns or patches.....	x	-	-
	Needle chews.....	x	-	-
	Box stitching less than specified length by more than 1/4 inch (except buckle end).....	-	-	x
	Box stitching at buckle end less than specified length by more than 1/8 inch.....	-	-	x
	Any component part of case spliced, i.e., constructed with more than one piece of duck or webbing (unless otherwise specified.).....	x	-	-
Binding	Loosely applied or badly puckered; end of binding not finished as specified.....	-	*	-
Snap fasteners (male or female)	Fail to snap closed or to remain in a closed position.....	x	-	-
	Male or female portion misplaced, failing to effect a smooth and secure closure.....	-	-	x
	Loosely clinched, causing male part to rotate in hole in fabric.....	-	-	x
	Clinched excessively tight, cutting surrounding fabric....	x	-	-
Clip ends	Improperly clinched, i.e., webbing not flat at point of entrance into clip.....	-	*	-
Flaps (2 required	Corners of flap irregularly shaped, or radius not uniform on both corners.....	-	-	x
Securing straps A & B	Strap reversed, affecting serviceability or appearance..	-	*	-
	Any buckle reversed, failing to effect a secure engagement of strap.....	x	-	-
Reinforcements	Any reinforcement improperly applied, causing excessive fullness on reinforcement or body of case.....	-	*	-

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4.3.3.1.2 Examination for dimensional characteristics.-

Examine	Defects	Classification	
		Major	Minor
Dimensions	Smaller than nominal dimensions less applicable minus tolerances indicated on drawing, but not smaller than nominal dimensions less twice the applicable minus tolerances.....		x
	Smaller than nominal dimensions less twice the applicable minus tolerances indicated on drawing.	x	
	Larger than nominal dimensions and applicable plus tolerances indicated on drawing.....		x

4.3.3.1.3 Inspection levels and acceptable quality levels (AQL's).- The inspection levels and the AQL's expressed in defects per hundred units shall be as follows:

	Inspection level	AQL	
		Major	Total
For defects applicable to 4.3.3.1.1	II	4.0	15.0
For defects applicable to 4.3.3.1.2	S-3	6.5	15.0

4.3.4 Examination of count of cases per intermediate container.- Cases packaged for shipment shall be examined to determine conformance with package markings and specified quantity. The sample unit for this examination shall be one box (interior package). Any box containing less than the specified or marked quantity of cases shall be classified as a defect. The inspection level shall be S-1 and the AQL shall be 4.0 defects per hundred units.

4.3.5 Examination of preparation for delivery requirements.- An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per 100 units.

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Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, sequence, location, or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping, loose strapping. Bulging or distortion of containers.
Weight and content	Number of intermediate packages is more or less than required; net weight exceeds requirements.

4.4 Tests.- The methods of testing specified in CCC-T-191 shall be followed wherever applicable.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Unit packaging.- Each case shall be neatly folded, straps fastened and packaged in a snug-fitting transparent polyethylene bag. The polyethylene bag shall be 0.004 inch thick. All seams and closures shall be made by heat sealing. Prior to or during the sealing operation, excess air shall be expelled.

5.1.1.2 Intermediate packaging.- Ten (10) cases prepared as specified in 5.1.1.1 shall be placed flat in a snug-fitting fiberboard box conforming to style RSC, type CF, class weather-resistant, grade W5C of PPP-B-636. Closure shall be in accordance with the appendix of the container specification.

5.1.2 Level C.- Cases shall be packaged to afford adequate protection against deterioration and damage during shipment from the supply source to the first receiving station for immediate use or controlled humidity storage.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.- Forty (40) cases (4 intermediate containers) packaged as specified in 5.1 shall be packed in a shipping container conforming to overseas type, style A or B or PPP-B-601 or class 2 of PPP-B-621. Closure shall be in accordance with the requirements of the applicable container specification.

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5.2.2 Level B.- Cases shall be packed as specified in 5.2.1, except the shipping container shall conform to domestic type of PPP-B-601 or class 1 of PPP-B-621.

5.2.3 Level C.- Cases packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rates available for such supplies. Shipping containers shall be in accordance with the rules and regulations of the carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, unit packages, interior package and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- Cases covered by this specification are intended to house the components of panel marker sets AP-30-C and AP-30-D.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When standard sample will not be furnished (see 3.1).
(c) When a preproduction sample is not required (see 3.2).
(d) Selection of applicable levels of packaging and packing (see 5.1 or 5.2).

6.3 Standard sample.- For standard samples (see 3.1) address the procuring office issuing the invitation for bids or request for proposal.

6.4 Weight.- The case covered by this specification will weight approximately 3.5 pounds.

Custodians:
Army - IH
Navy - SA
Air Force - 82

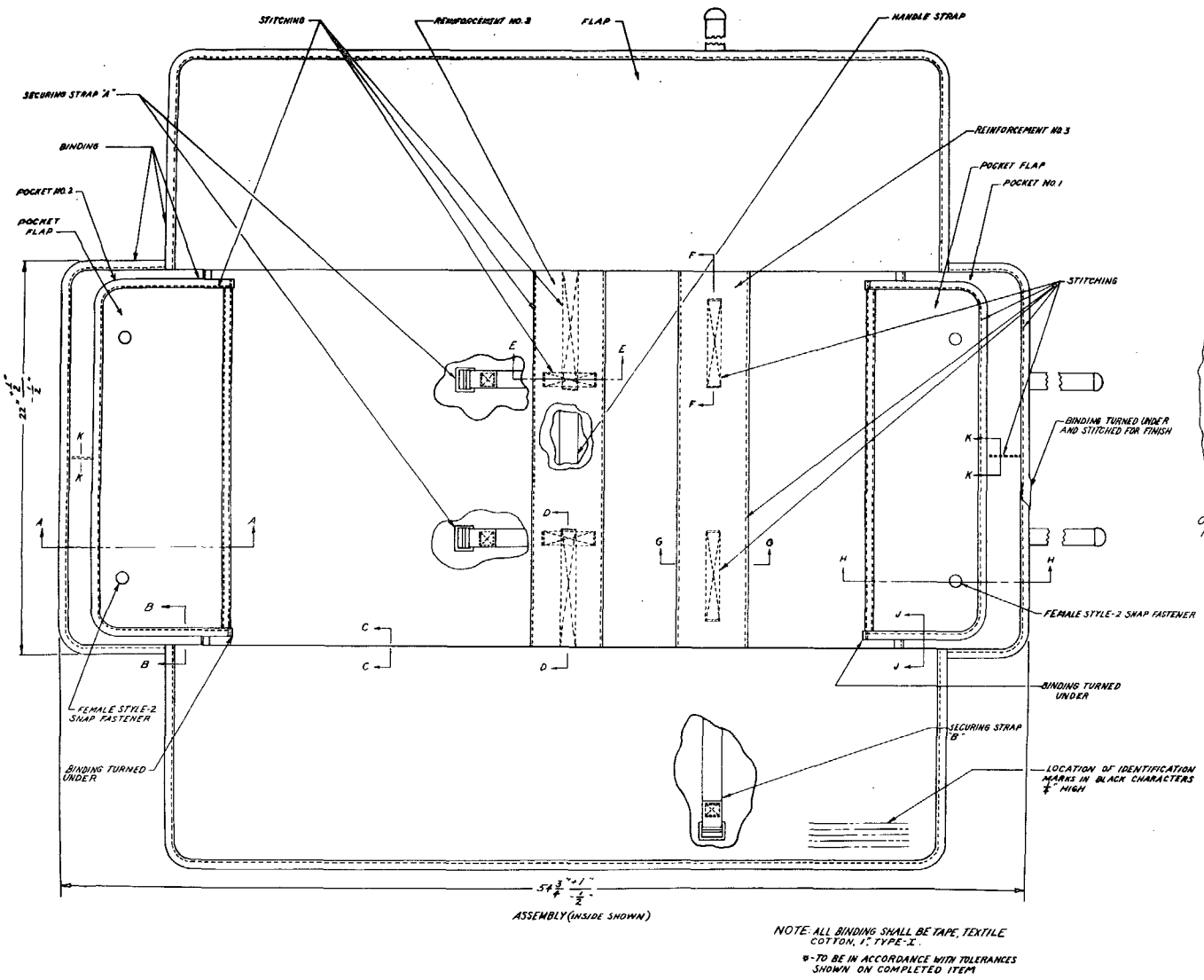
User Activities:
Navy - MC
Army - SM

Preparing Activity:
The Institute of Heraldry, United States Army

Review Activities:
Navy - SA
Air Force - 82, 85
Other - DP

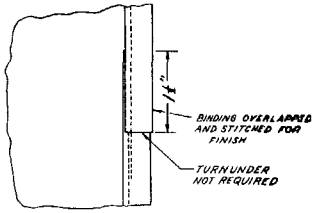
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ILLUSTRATION



OPTIONAL METHOD OF FINISHING BINDING

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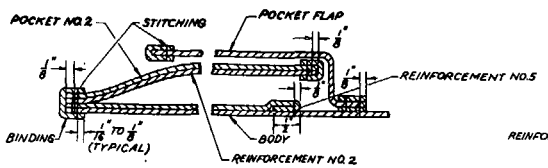
NOTE: ALL BINDING SHALL BE TAPE, TEXTILE COTTON, 1" TYPE-X.

* - TO BE IN ACCORDANCE WITH TOLERANCES SHOWN ON COMPLETED ITEM

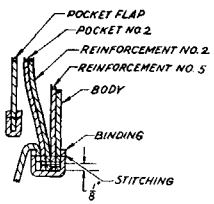
CASE, PANEL MARKER FOR ADVANCE COMMUNICATION PANEL MARKER SETS AP-30-C AND AP-30-D
ASSEMBLY AND ILLUSTRATION

FIGURE 1

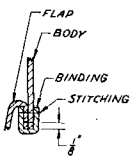
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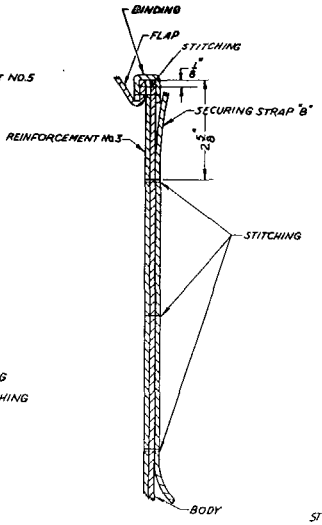
SECTION A-A



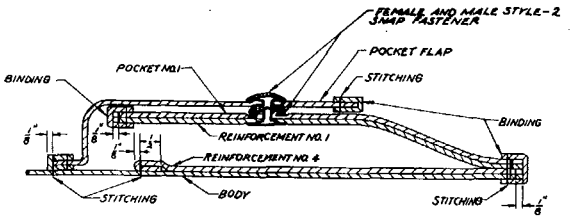
SECTION B-B



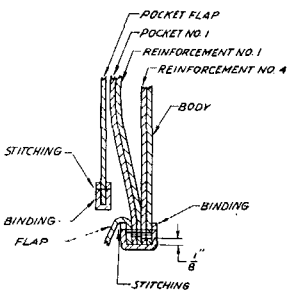
SECTION C-C



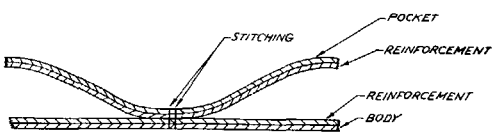
SECTION F-F



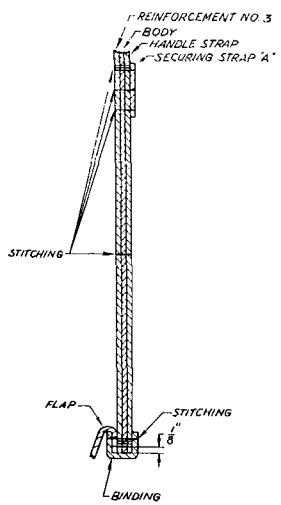
SECTION H-H



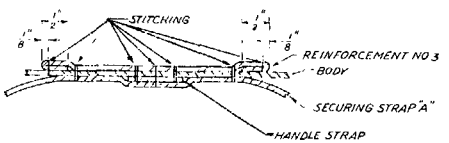
SECTION J-J



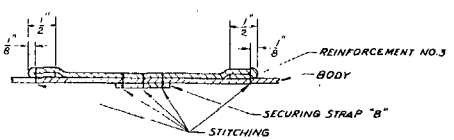
SECTION K-K



SECTION D-D



SECTION E-E



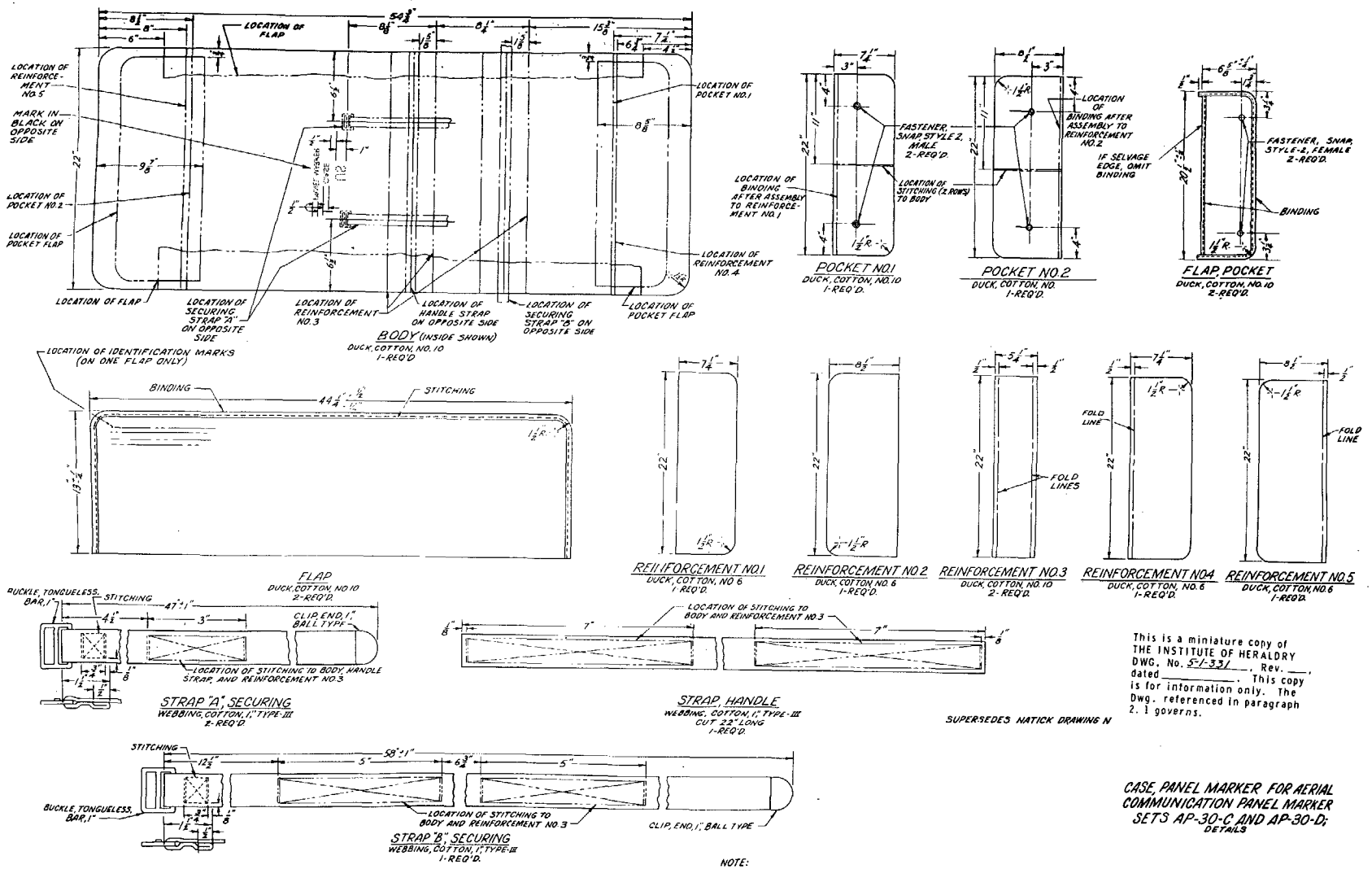
SECTION G-G

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CASE, PANEL MARKER FOR AERIAL COMMUNICATION PANEL MARKER SETS AP-30-C AND AP-30-D SECTIONS

FIGURE 2

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SUPERSEDES NATICK DRAWING N

NOTE:
A TOLERANCE OF ± 1/8-INCH WILL BE ALLOWED ON THE LOCATION DIMENSIONS OF THE FOLLOWING COMPONENTS:
A. STRAP, HANDLE
B. FLAPS
C. STRAP, SECURING, "A" & "B"

CASE, PANEL MARKER FOR AERIAL COMMUNICATION PANEL MARKER SETS AP-30-C AND AP-30-D; DETAILS

FIGURE 3

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